Dylan Johnson Engineering Project Portfolio

B.S. Mechanical Engineering | CU Boulder 2025

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Mechanical Engineering graduate eager to apply project coordination, stakeholder communication, and technical problem-solving skills to support NASSCO's mission of securing our nation and fueling our economy.

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TB2 Aerospace Fuel Pod – Senior Capstone Project

Role: CAD & Fluids Engineer

Objective: Design stackable, autonomously deployable composite fuel pods aimed at reducing soldier exposure and saving lives by minimizing time spent at Forward Arming and Refueling Points (FARPs).

Figure 1: Fleet of six pods refueling a Black Hawk, stacked as they would be in the field.

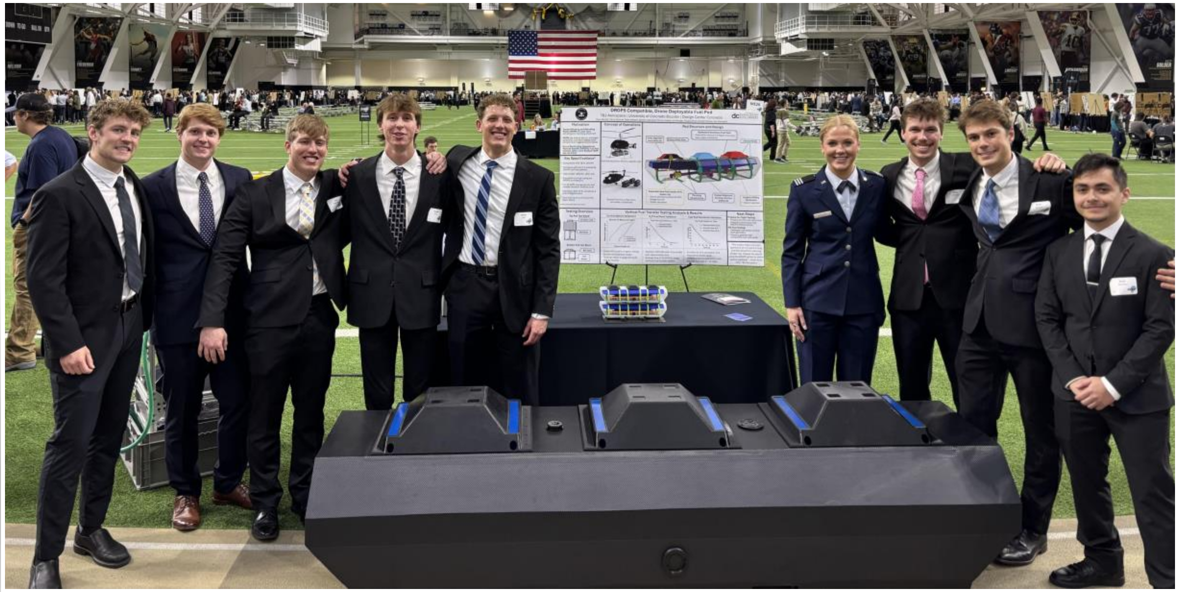


Figure 2: TB2 capstone team in front of the full-scale pod at the 2025 Senior Design Expo.

TB2 Aerospace Fuel Pod – Structure and Design

- Designed and modeled UAV-deployable fuel pods in SolidWorks, enabling automated forward refueling and reducing soldier exposure by 75%.
- Led Vertical Fuel Transfer (VFT) subsystem CAD development, cutting refueling connections by 50% (detailed on next slide).
- Designed composite panel outer shell with design for manufacturability and assembly in mind, leading to manufacturing partner approval by QuickStep.
- Managed project BOM and engineering change requests using BILD PDM, tracking all modifications for team alignment.

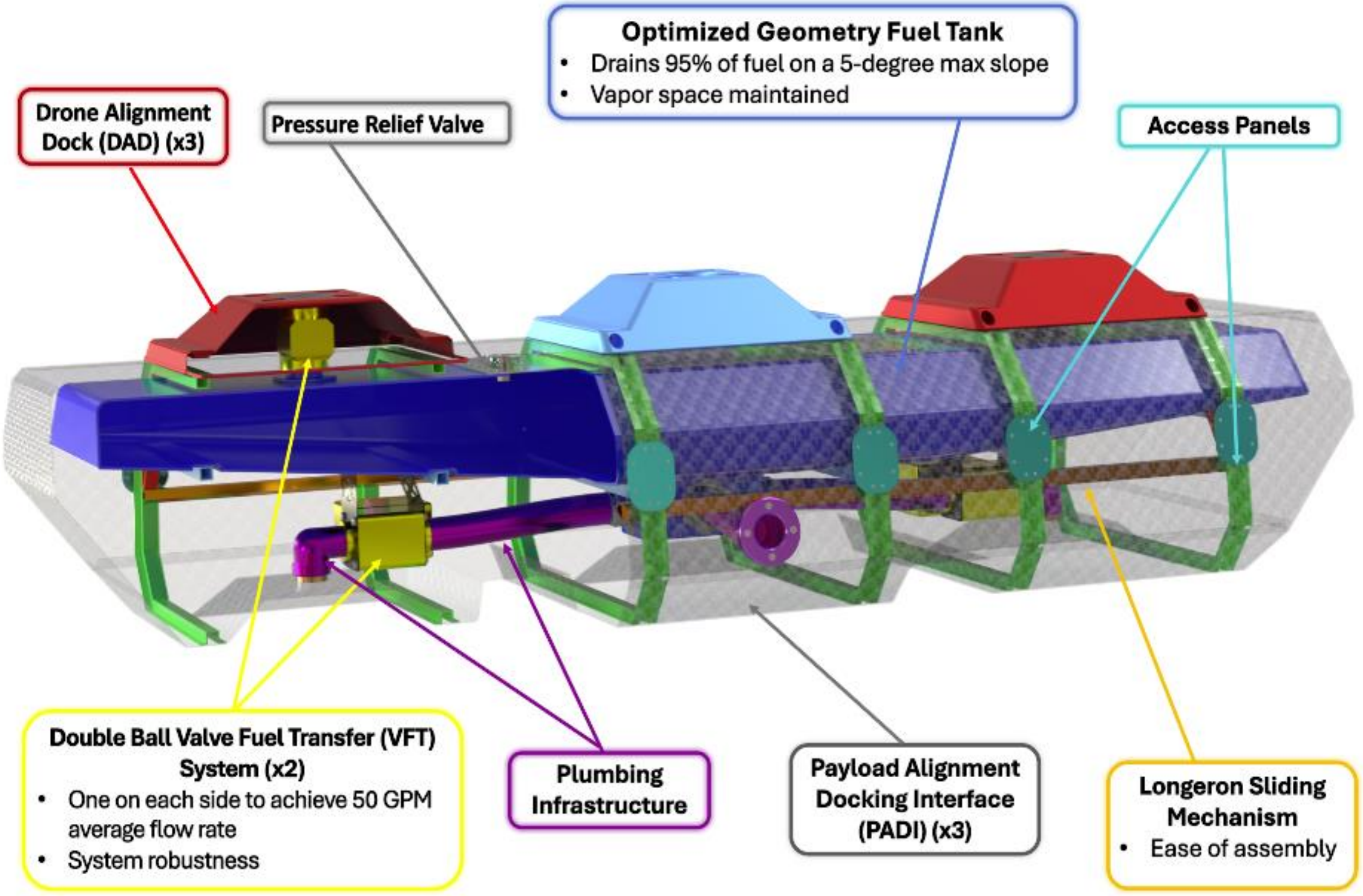


Figure 3: Internal pod structure with critical system architecture highlighted.

TB2 Aerospace Fuel Pod – Vertical Fuel Transfer

Goal: Automatically transfer fuel between stacked pods at an average flow rate greater than 50 GPM.

Initial Design Concept: Strictly mechanical, stacking force initiates fuel transfer.

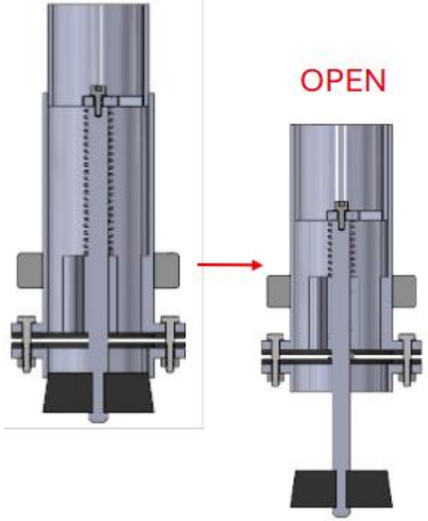


Figure 4: Original telescoping mechanical design.

- Nozzle Protrusion Increased Failure Risks
- Custom and OTS Components

Final Design Concept: Electronic ball valves initiate and control flow via RF communication upon stacking.

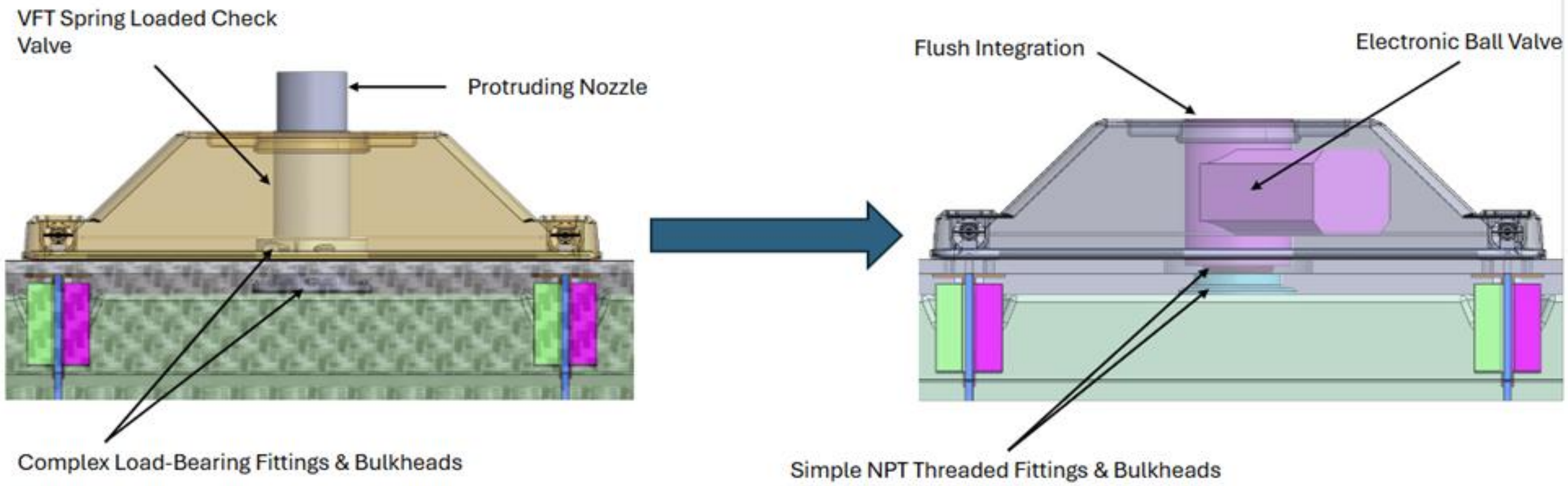


Figure 5: Final design utilizing electronic ball valves to mitigate flow; purely fuel-grade off the shelf components.

- BOM count reduced by 42 parts
- All OTS components significantly reduced costs
- Mechanical override solution presented along with financial and BOM to justify decisions to client

TB2 Aerospace Fuel Pod – Testing and Results

- Designed, procured, and assembled parts for a subsystem test rig.
- Wrote a MATLAB model using differential equations to predict height of fuel in tank over time and matched drain time within 1% of experimental results.
- Confirmed model accuracy with test rig geometry as inputs and applied it to the custom tank to confirm flow rate specs were met.

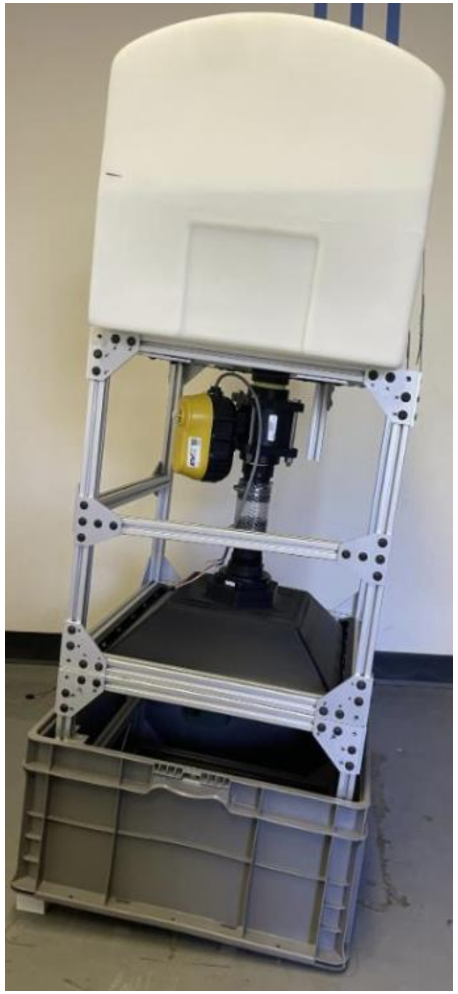
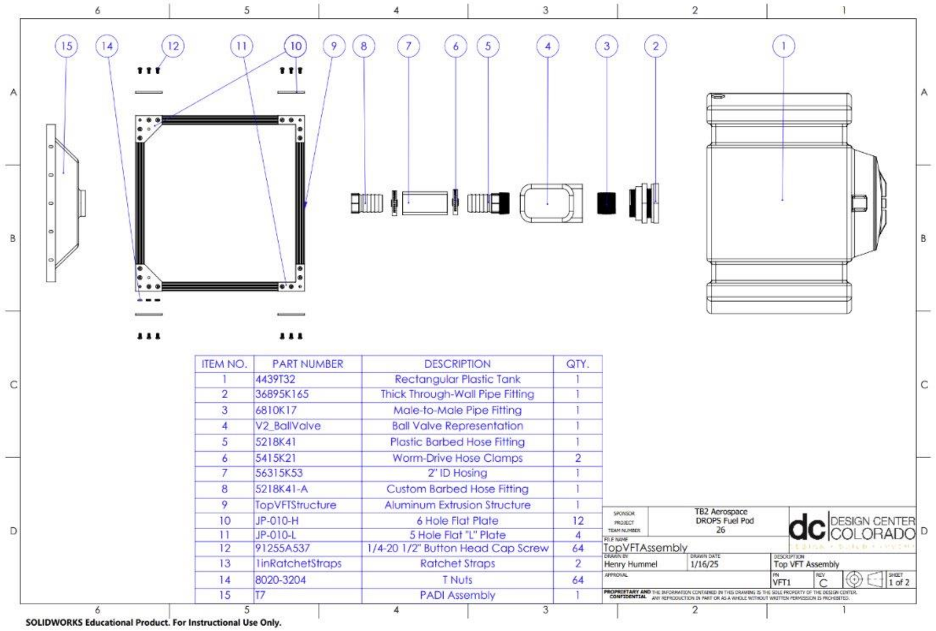


Figure 6: Vertical Fuel Transfer test rig assembly.



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	4439T32	Rectangular Plastic Tank	1
2	36895K165	Thick Through-Wall Pipe Fitting	1
3	6810K17	Male-to-Male Pipe Fitting	1
4	V2_BallValve	Ball Valve Representation	1
5	5218K41	Plastic Barbed Hose Fitting	1
6	5415K21	Worm-Drive Hose Clamps	2
7	56315K53	2" ID Hosing	1
8	5218K41-A	Custom Barbed Hose Fitting	1
9	TopVFTStructure	Aluminum Extrusion Structure	1
10	JP-010-H	6 Hole Flat Plate	12
11	JP-010-L	5 Hole Flat "L" Plate	4
12	91255A537	1/4-20 1/2" Button Head Cap Screw	64
13	1inRatchetStraps	Ratchet Straps	2
14	8020-3204	T Nuts	64
15	T7	PADI Assembly	1

Figure 7: Assembly drawing with BOM for top half of test rig (full rig is a stacked top and bottom).

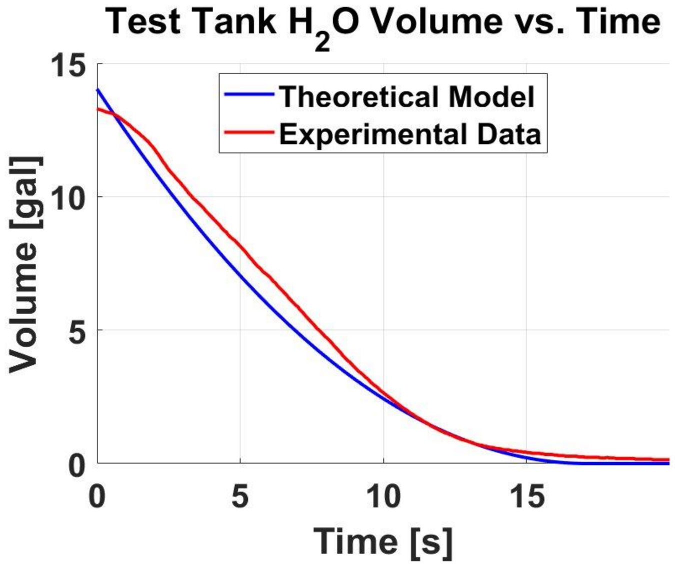


Figure 8: Theoretical model vs. experimental data comparison as plotted on MATLAB.

TB2 Aerospace Fuel Pods – Stakeholder Presentations & Critical Reviews

- Collaborated with U.S. Army personnel at Boulder Airport to discuss CONOPS for Black Hawk operations.
- Presented to UK Ministry of Defense Strategic Command as part of a select team, delivering high-level project briefings.
- Facilitated cross-company collaboration by briefing Stratom Inc.'s chief engineer alongside TB2 leadership.
- Prepared and delivered design & manufacturing reviews (PDR, CDR, TRR) to the client, defending scope and confirming project specs were met.

Figure 9: U.S. Army flies Black Hawk to Boulder Municipal Airport to collaborate with TB2.

Drill Powered Trike Project

Role: CAD & Manufacturing Engineer

Objective: Design and fabricate a 3+ wheel vehicle powered solely by a Ryobi drill to complete a maneuverability course, staying under a 50 lb. weight limit and $200 budget.

Figure 12: Machined drill-plate assembly mounted to the front fork of the trike.

- Designed and analyzed a custom trike frame using structural analysis and FEA, ensuring strength and performance under load.
- Managed procurement and vendor coordination for steel, gears, bearings, and PLA filament, optimizing cost and availability.
- Fabricated custom parts via MIG welding, milling, turning, 3D printing, and water jet cutting.

Figure 10: Custom designed and MIG welded trike frame.

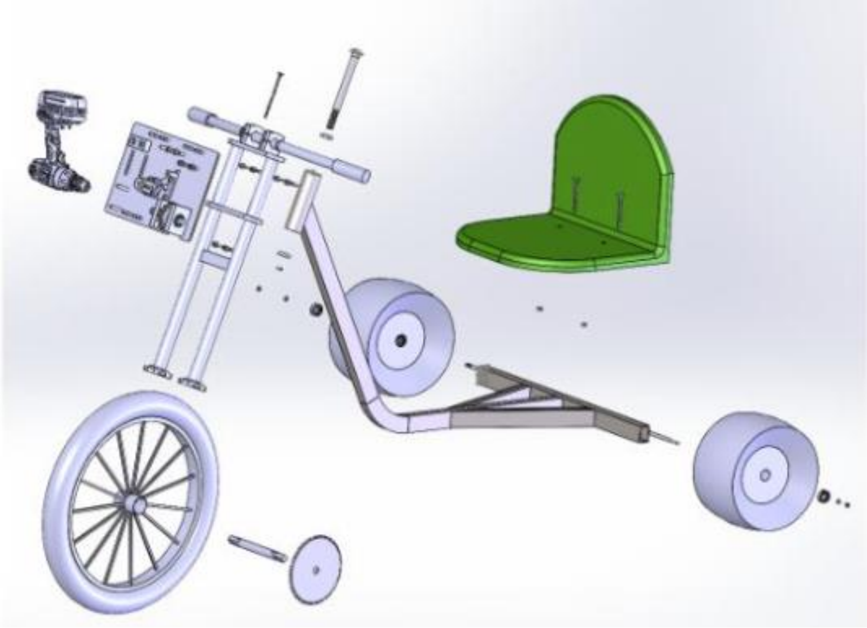


Figure 11: Exploded view of the top-level SolidWorks assembly.

Drill-Powered Trike – Drive Train Assembly

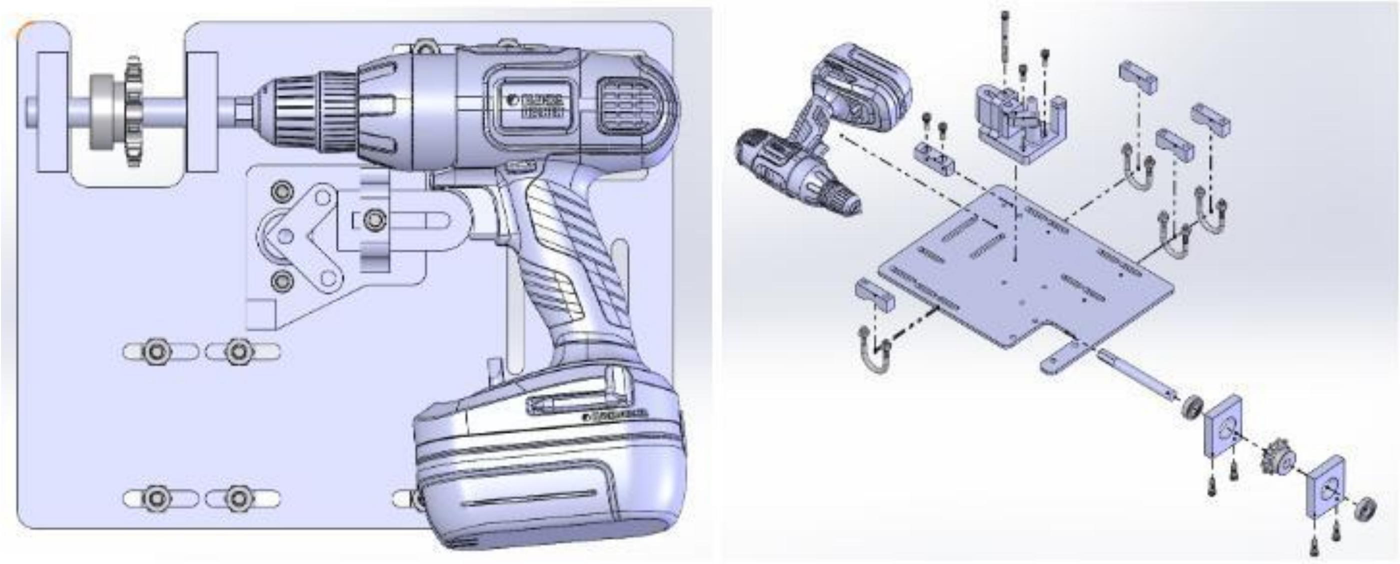


Figure 13: Drill mounting plate assembly and exploded view.

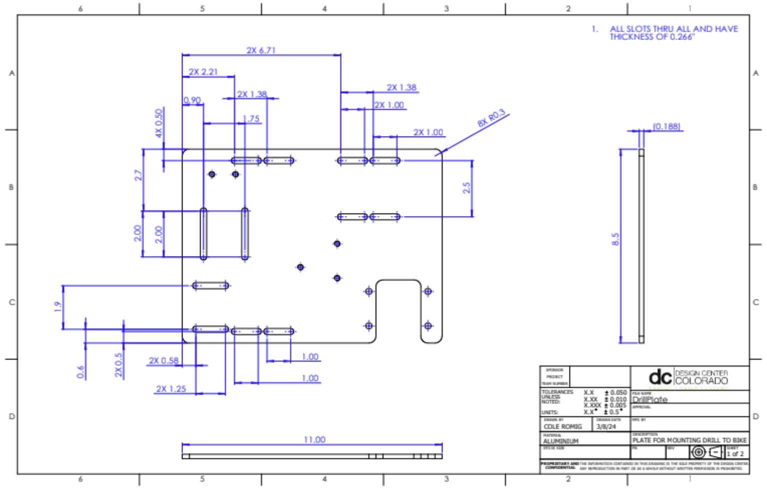


Figure 14: Drill-plate drawing 1/2.

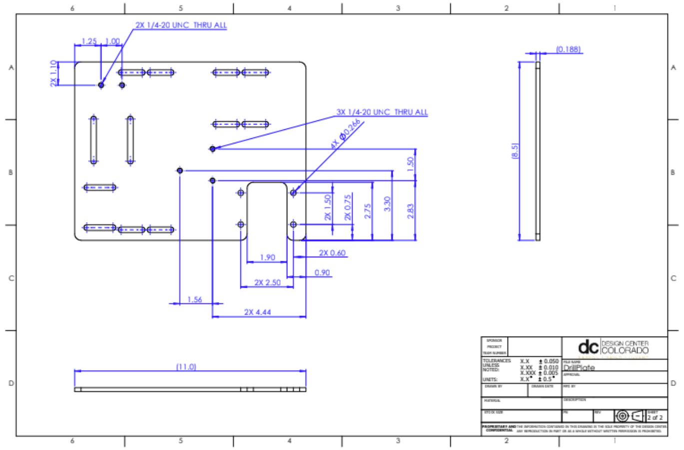


Figure 15: Drill-plate drawing 2/2.

- Modeled and produced detailed drawings for a drill-plate assembly, including a gear train driven by press-fit ball bearings.
- Engineered a custom activation mechanism with a 3D-printed trigger attachment, tensioned via a break line for precise actuation.

FormFit Solutions - Entrepreneurship Contest

Role: Project Manager/CEO

Objective: Identify a common problem and develop an economically and technically viable solution to compete in the 2024 Integrated Design Engineering (IDE) Expo.

- Led team to 1st place at the 2024 IDE Entrepreneurship Competition.
- Managed project schedule, budget, and BOM, ensuring critical deadlines were met and materials procured.
- Directed team meetings and coordinated rapid prototyping using 3D printing, woodworking, and laser cutting.
- Presented to a judge panel, effectively communicating design intent and financial projections.

Figure 16: Proposed product to solve the problem of inefficient and expensive tool organization.

Wobbler Engine – CAD & Machining Project

Objective: Collaborate with three other engineering students to model and fabricate a Compressed-air powered wobbler engine assembly.

- Collaborated with three other engineering students to model, draft, and machine a compressed-air powered wobbler engine using SolidWorks and CU's machine shop.
- Gained hands-on machining experience with mills and lathes.
- Assembled a functional engine with proper tolerancing and five custom components.

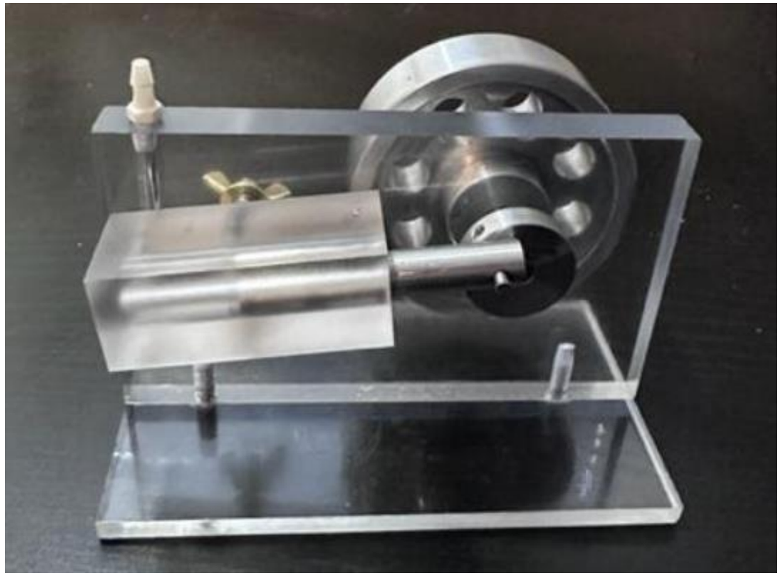
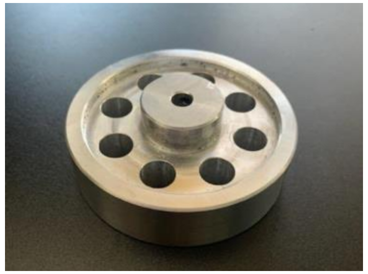


Figure 18: Wobbler engine final assembly.



Figures 19 and 20: Fly-wheel and crank disk components machined via turning and 3-axis mill.

Figure 17: Valve plate as machined via 3 axis mill.

Tactic Air Drone - Reverse Engineering and Modeling

Objective: Reverse engineer and model a real-world, complex assembly on SolidWorks.

Figure 21: Real Drone in folded mode.

- Reverse-engineered and modeled a tactical foldable drone in SolidWorks using calipers to capture critical dimensions.
- Created detailed part and assembly drawings for manufacturing.
- Performed interference detection to ensure proper folding and dimensional accuracy.

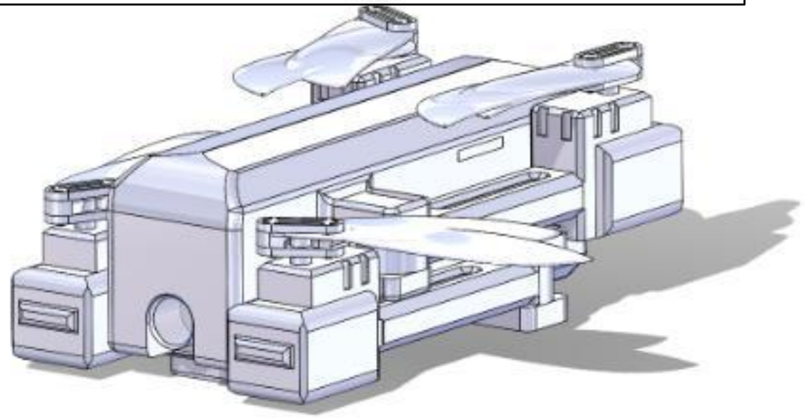


Figure 22: Drone in folded mode as modeled in SolidWorks.

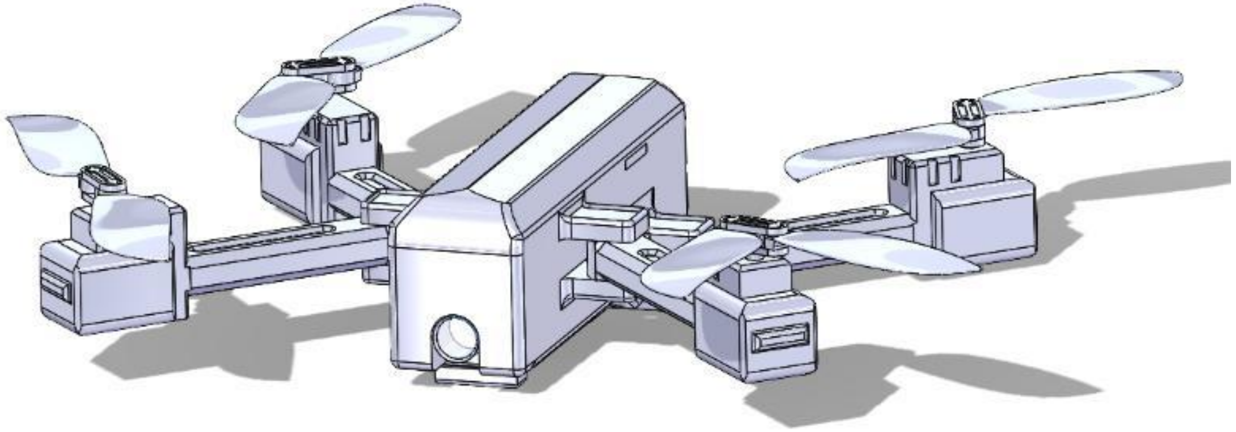


Figure 23: Drone in flight mode as modeled in SolidWorks.

My Dream: See the World

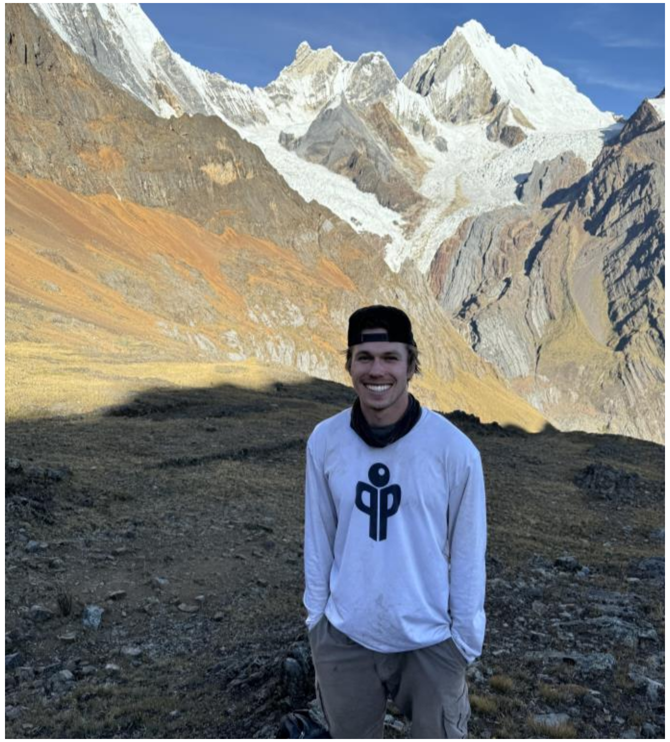


Figure 24: Cordillera Huayhuash Trek: Andes Mountains, Peru.

Max Elevation: Paso Yaucha, 16,200 ft

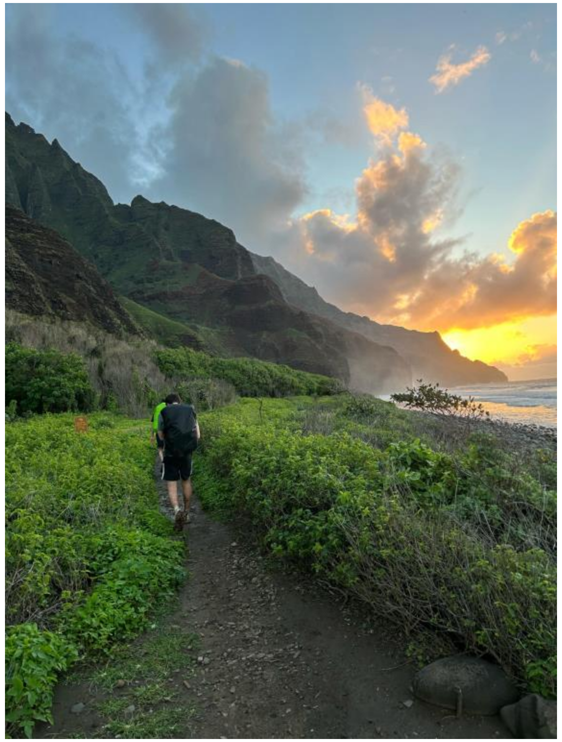
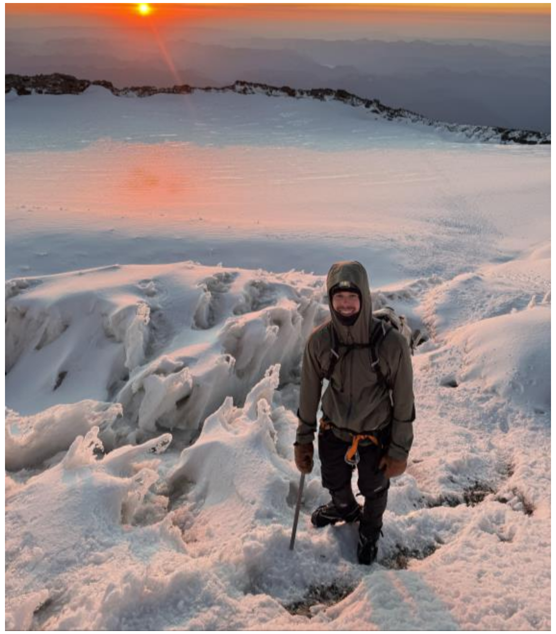


Figure 25: Nepali Coast: Kuai, Hawaii.

Round Trip: 22 Miles
Elevation Gain: 7,000 ft



Figures 26 & 27: Mt. Ranier Summit day: 8/23/25. Max Elevation: 14,410 ft.

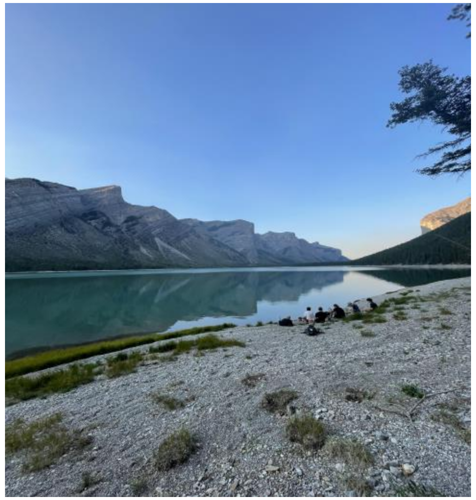


Figure 28: Ghost Lakes: Banff, Canada.

Figure 29: Haleakala Crater: Maui, Hawaii.